

1.

(a) Natural number: 5

(b) Integer that is not natural: -2

(c) Rational number that is not integer: $\frac{1}{2}$ (d) Irrational number $\sqrt{2}$

$$2. (a) \quad 3.14 \times 10^7 = \cancel{314,000,000} \\ \boxed{314,000,000} \\ \cancel{31,400,000}$$

$$(b) \quad 2.764 \times 10^6 = \cancel{2,764,000} \quad \boxed{2,764,000}$$

$$(c) \quad 2.61 \times 10^{-3} = \cancel{0.00261} \quad \boxed{0.00261}$$

$$(d) \quad 9.999 \times 10^{-2} = \cancel{0.09999} \quad \boxed{0.09999}$$

$$3. \quad x^3 + 10x^2 + 25x \\ = x(x^2 + 10x + 25) \\ = x(x+5)^2$$

$$\text{Ans: } \boxed{x(x+5)^2}$$

(2)

$$\begin{aligned}
 4. \quad \frac{9}{2-\sqrt{5}} &= \frac{9}{2-\sqrt{5}} \cdot \frac{2+\sqrt{5}}{2+\sqrt{5}} \\
 &= \frac{9(2+\sqrt{5})}{\cancel{4-5} \quad 4+5} = \frac{9(2+\sqrt{5})}{9} \\
 &= \frac{\cancel{9(2+\sqrt{5})}}{\cancel{9}} = \cancel{2+\sqrt{5}} \\
 &\quad 2+\sqrt{5}
 \end{aligned}$$

~~Ans:~~Ans: $\boxed{2+\sqrt{5}}$

5. Let width be w , length be $w+5$
 since diagonal is 145

$$w + (w+5) = 145$$

$$2w + 5 = 145$$

$$2w = 140 \quad w = 70, l = 75$$

Ans: $\boxed{\text{width} = 70, \text{length} = 75}$

(3)

6. (a) $9x^2 - 25y^2 = 225$

For x -intercepts, set $x=0$

$$-25y^2 = 225, y^2 = -9$$

So x -intercepts

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For y -intercepts, set $y=0$

$$9x^2 = 225, x = \pm 5$$

So y -intercepts: $(-5, 0), (5, 0)$

(b) $y^2 - 2xy + 5x = 1$

For x -intercept, set $x=0$

$$y^2 = 1, y = \pm 1$$

So x -intercepts: $(0, 1), (0, -1)$

For y -intercepts, set $y=0$

$$5x = 1, x = \frac{1}{5}$$

So, y -intercepts: $(\frac{1}{5}, 0)$

7.

P varies inversely as S

$$P = \frac{k}{S}$$

Ans: $\boxed{P = \frac{k}{S}}$

8. $f(x) = 3x - 2$, $g(x) = 4 - x^2$

(a) $(f \circ f)(-1) = f(f(-1))$

$$f(-1) = 3(-1) - 2 = -5$$

$$f(-5) = 3(-5) - 2 = -17$$

Ans: $\boxed{-17}$

(b) $(g \circ g)(-1) = g(g(-1))$

$$g(-1) = 4 - (-1)^2 = 3$$

$$g(3) = 4 - 3^2 = -5$$

Ans: $\boxed{-5}$

(4)

9. $f(x) = -x + 5$, $f(-5) = 10$

$f(-4) = 9$, $f(0) = 5$

$f(1) = 4$, $f(4) = 1$, $f(5) = 0$

Table values: $\boxed{10, 9, 5, 4, 1, 0}$

Graph: Line segment from $(-5, 10)$ to $(5, 0)$

10. $f(x) = |3x - 3|$, $f(-5) = 18$

$f(-2) = 9$, $f(0) = 3$

$f(1) = 0$, $f(2) = 3$, $f(5) = 12$

Table values: $\boxed{18, 9, 3, 0, 3, 12}$

The graph is a V-shape, so I choose the upward V with vertex at $(0, 0)$

Ans: $\boxed{\text{bottom-left graph}}$

11.

$$r(x) = \frac{3x-4}{x^2-16}$$

$$x^2-16 = (x-4)(x+4)$$

since the denominator becomes zero;

$$x = 4, -4$$

Horizontal asymptotes: $x=4, x=-4$

since the degree of the denominator is bigger:

$$y=0$$

vertical asymptote: $y=0$

12.

$$\log x = -5, x = 10^{-5}$$

Ans: 10^{-5}

13.

$\sin(\sin^{-1}(\frac{1}{8}))$, using calculator,

$$\sin^{-1}(\frac{1}{8}) \approx 0.1253$$

$$\sin(0.1253) \approx 0.12497$$

Ans: 0.125

14.

$$\frac{\sin \theta}{56.3} = \frac{\sin 66^\circ}{80.2}$$

$$\sin \theta = \frac{56.3 \sin 66^\circ}{80.2}$$

$$\sin \theta = \frac{56.3 (0.9135)}{80.2}$$

$$\sin \theta = \frac{51.44}{80.2}, \quad \sin \theta = 0.614$$

$$\theta = \sin^{-1}(0.614), \quad \theta = 37.9^\circ$$

$$\text{Ans: } \boxed{37.9^\circ}$$

$$15. \quad y = 6x^2, \quad x = r \sin \theta, \quad y = r \cos \theta$$

$$r \cos \theta = 6(r \sin \theta)^2$$

$$r \cos \theta = 6r^2 \sin^2 \theta, \quad \cos \theta = 6r \sin^2 \theta$$

$$r = \frac{\cos \theta}{6 \sin^2 \theta}$$

$$\text{Ans: } \boxed{r = \frac{\cos \theta}{6 \sin^2 \theta}}$$

16. major axis length 8, so;

$$2a = 8, a = 4$$

minor axis length 6, so:

$$2b = 6, b = 3$$

Foci on x-axis means horizontal ellipse

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1, \quad \frac{x^2}{16} + \frac{y^2}{9} = 1$$

Ans:

$$\boxed{\frac{x^2}{16} + \frac{y^2}{9} = 1}$$

17.